Medication Policy Manual

Policy No: dru672

Topic: Medications for Multiple Myeloma, other cancers, and other hematologic disorders

Date of Origin: October 1, 2021

- Darzalex, daratumumab
- Darzalex Faspro, daratumumab and hyaluronidase-fihj
- Elrexfio (elranatamab-bcmm)
- Empliciti, elotuzumab
- Kyprolis, carfilzomib
- Ninlaro, ixazomib
- Pomalyst, pomalidomide
- Sarclisa, isatuximab-irfc
- Talvey (talquetamab-tgvs)
- Tecvayli, teclistamab-cqyv

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IMPORTANT REMINDER

This Medication Policy has been developed through consideration of medical necessity, generally accepted standards of medical practice, and review of medical literature and government approval status.

Benefit determinations should be based in all cases on the applicable contract language. To the extent there are any conflicts between these guidelines and the contract language, the contract language will control.

The purpose of Medication Policy is to provide a guide to coverage. Medication Policy is not intended to dictate to providers how to practice medicine. Providers are expected to exercise their medical judgment in providing the most appropriate care.

Description

Medications included in this policy are used primarily to treat multiple myeloma, but also include various other cancers, and other hematologic disorders.

Policy/Criteria

Most contracts require pre-authorization approval of Medications for Multiple Myeloma, other cancers, and other hematologic disorders ("Medications for MM") prior to coverage.

I. Continuation of therapy (COT): Medications for MM may be considered medically necessary for COT when criteria A and B below are met.

A. The patient is established on this therapy AND one of the following situations applies (criterion 1 or 2 below):

1. Prior to current health plan membership AND the medication was covered by another health plan.

PLEASE NOTE: *If the diagnosis is not listed in the coverage criteria below, written documentation of coverage must be provided, such as an approval letter or paid claim.*

OR

2. The medication was initiated for acute disease management, as part of an acute unscheduled, inpatient hospital admission.

AND

B. If the diagnosis is listed in the 'Not Medically Necessary Uses' or 'Investigational Uses' coverage criteria below, documentation of clinical benefit, such as disease stability as detailed in the reauthorization criteria must be provided.

Please note*: Medications obtained as samples, coupons, or promotions, paying cash for a prescription ("out-of-pocket") as an eligible patient, or any other method of obtaining medications outside of an established health plan benefit (from your insurance) does NOT necessarily establish medical necessity. Medication policy criteria apply for coverage, per the terms of the member contract with the health plan.*

II. New starts (treatment-naïve) patients – Medically Necessary Uses:

Medications for multiple myeloma (MM) (as listed in Table 1) may be considered medically necessary when there is clinical documentation (including, but not limited to chart notes) that the applicable diagnosis-based criteria below are met.

Table 1.

Diagnosis	Coverable medication(s)	AND the following are met:
Kaposi sarcoma (KS)	- Pomalyst (pomalidomide)	**1.** Prior chemotherapy (such as listed in *Appendix 1*) was not effective, is contraindicated, or was not tolerated. **AND** **2.** If the KS is associated with acquired immune deficiency syndrome (AIDS-related), highly active antiretroviral therapy (HAART) was not effective.
Multiple myeloma (MM)	- Kyprolis (carfilzomib) - Darzalex (daratumumab) - Darzalex Faspro (daratumumab and hyaluronidase-fihj) - Elrexfio (elranatamab) - Empliciti (elotuzumab) - Ninlaro (ixazomib) - Pomalyst (pomalidomide) - Sarclisa (isatuximab-irfc) - Talvey (talquetamab) - Tecvayli (teclistamab-cqyv)	**1.** A diagnosis of MM. **AND** **2.** ***For Darzalex, Darzalex Faspro, Empliciti, Sarclisa only***: Will <u>not</u> be used in combination with another monoclonal antibody [such as listed in *Appendix 1*]. **AND** **3.** ***For Empliciti, Sarclisa, Ninlaro only:*** The MM is relapsed or refractory to at least one prior therapy. **AND** **4.** ***For Sarclisa only*****:** The MM was <u>not</u> refractory to prior daratumumab *(refractory is defined as disease progression while on therapy, or within 60 days of the last dose).* **AND** **5.** ***For Tecvayli, Elrexfio, and Talvey only*****:** Criteria a, b, and c below are met: **a.** There has been disease progression on or after at least <u>four</u> prior MM regimens including <u>all</u> of the following (i, ii, and iii): **i.** An anti-CD38 monoclonal antibody. **ii.** A proteosome inhibitor. **iii.** An immunomodulatory (IMID) agent. **b.** Tecvayli, Elrexfio, or Talvey will be used as monotherapy. **c.** For Tecvayli and Elrexfio only: No prior treatment with a therapy directed against B-cell maturation antigen (BCMA). (refer to *Appendix 1*)
Light chain amyloidosis (AL)	- Darzalex (daratumumab) - Darzalex Faspro (daratumumab and hyaluronidase-fihj)	**1.** A diagnosis of light chain amyloidosis (AL). **AND** **2.** The patient has had no prior therapy for AL (newly diagnosed). **AND** **3.** Daratumumab will be administered in combination with bortezomib, cyclophosphamide, and dexamethasone. **AND** **4.** No prior treatment with daratumumab.

III. Administration, Quantity Limitations, and Authorization Period

A. Asuris Pharmacy Services considers oral medications [including Ninlaro (ixazomib), Pomalyst (pomalidomide)] coverable only under the pharmacy benefit (as self-administered medications).

B. Asuris Pharmacy Services considers injectable medications [including Kyprolis (carfilzomib), Darzalex (daratumumab), Darzalex Faspro (daratumumab and hyaluronidase-fihj), Empliciti (elotuzumab), Elrexfio (elranatamab-bcmm), Sarclisa (isatuximab-irfc), Talvey (talquetamab-tgvs) and Tecvayli (teclistamab-cqyv)] coverable only under the medical benefit (as provider-administered medications).

C. When pre-authorization is approved, each drug will be authorized as follows:

1. **Self-administered medications:** Up to the limits in Table 2 until disease progression.
2. **Provider-administered medications:** Up to FDA-recommended dose and frequency limits until disease progression.

Table 2. Self-Administered Medication Quantity Limits

Product	Quantity Limit
Ninlaro (ixazomib)	3 capsules per 28 days.
Pomalyst (pomalidomide)	- **MM:** Up to 21 capsules per 28 days. - **KS:** Up to 42 capsules per 28 days.

Key: KS=Kaposi sarcoma; MM=multiple myeloma

D. Authorization **may** be reviewed at least annually. For any ongoing authorization, clinical documentation (including, but not limited to chart notes) must be provided to confirm that current medical necessity criteria are met, and that the medication is providing clinical benefit, such as disease stability or improvement.

IV. Investigational Uses

A. Combination use of any medications in this policy, if specifically excluded in the coverage criteria above, including use of any two monoclonal antibodies for MM, including but not limited to Darzalex (daratumumab), Empliciti (elotuzumab), or Sarclisa (isatuximab-irfc) [see *Appendix 1*].

B. Unless otherwise specified in the coverage criteria above, medications included in this policy are considered investigational when used for all other conditions, due to lack of published data, lack of high-quality data, or lack of positive data.

Position Statement

Summary

- The intent of this policy is to cover medications for multiple myeloma, other cancers, and hematologic disorders ("medications for MM") in settings where they have been shown to be safe and effective, as detailed in the coverage criteria, with consideration for other available treatment options.
  * Where there is lack of proven additional benefit and/or lack of demonstrated health outcomes (such as overall survival or improved quality of life) relative to alternative therapies, use of medications for MM is not coverable ("not medically necessary" or "investigational").
  * It is important to note that the fact that a medication is FDA approved for a specific indication does not, in itself, make the treatment medically reasonable and necessary.
- Many of the clinical indications for medications for MM have been approved by the FDA and endorsed by clinical guidelines based on surrogate measures such as overall tumor response rate (ORR) and progression-free survival (PFS), measurements of tumor markers, which are not proven to accurately predict clinically important outcomes for MM, such as improved overall survival (OS), symptom control, or improved quality of life (QoL).
- Triplet MM regimens (drug regimens that combine medications from three different MM medication classes) have become the standard of care in treating MM. However, there are insufficient studies that compare regimens to clearly establish superiority of any one regimen or medication within a class (by mechanism of action).
- The safety and efficacy of monoclonal antibodies (mAbs) for MM in combination with other monoclonal antibodies for MM, or in conditions not included in coverage criteria (as listed above), have not been established. Currently, there are no published trials of the use of combination anti-MM monoclonal antibodies. Additional trials are ongoing.
- The National Comprehensive Cancer Network (NCCN) multiple myeloma guideline lists all drugs in this policy for use in MM in various settings, as well as other associated cancers and hematologic disorders. Choice of initial MM therapy is frequently based on transplant eligibility.
- Medications for MM are coverable for up to the dose and quantity as specified in the coverage criteria. For many medications for MM, they are given until disease progression, unacceptable toxicity, where others may be given for a specific duration (refer to coverage criteria). There is no conclusive additional benefit with higher doses or when given for longer durations except as specified in the coverage criteria.
- There are ongoing studies using medications for MM in a variety of other settings and other cancers. However, although initial evidence may be promising, the potential for clinical benefit in these conditions is still being investigated.

Asuris Pharmacy Services performs independent analyses of oncology medication evidence, to establish coverability per the contracts with the health plan.

- Most contracts define coverability based on established clinical benefit in published, peer-reviewed literature, along with consideration of regulatory status.
- FDA approval does not in itself establish medical necessity, as unpublished, low-quality evidence, including exploratory analyses and unvalidated surrogate endpoints, may be used as the basis of approval. Regulatory approval may or may not reflect clinical benefit relative to standard of care and the recommendations of expert clinical advisors such as the Oncologic Drugs Advisory Committee (ODAC). FDA approvals generally do not consider cost compared to established therapies, or value to members.
- Likewise, NCCN clinical practice guidelines assignment of a recommendation (category 1, 2a, or 2b) does not necessarily establish medically necessity. NCCN recommendations are inconsistently supported by published, peer-reviewed literature and do not uniformly consider value of new therapies relative to existing potentially higher-value treatment options, considering effectiveness, safety, and cost.
- Medication coverage criteria are developed based on the 'medical necessity' assessment, as described above.

Asuris Pharmacy Services analysis and coverage policy may differ from FDA labeled indication and/or NCCN clinical practice guidelines.

Clinical Efficacy

<u>Multiple Myeloma (MM)</u>

- Medications for multiple myeloma may be covered as detailed in the coverage criteria when there is documentation of a diagnosis of multiple myeloma.
- MM is a malignant neoplasm of plasma cells. The cells accumulate in bone marrow which leads to bone destruction and marrow failure. Skeletal destruction can lead to osteolytic lesions, osteopenia, and/or pathologic fractures. Bone pain is present at diagnosis in approximately 60% of patients. [1]
- MM is not curable using current approaches. Nearly all patients relapse on their initial treatment and require further therapy. The duration of response is generally shorter with each successive therapy. The five-year survival rate is approximately 52%. [2]
- Several subtypes of MM have been identified at the genetic and molecular level. Specific chromosomal translocations, deletions, and amplifications can be used to stratify disease risk (high, intermediate-, or standard-risk). [2]
- Choice of therapy may be based on several characteristics including whether the patient is a candidate for transplant, the disease risk category, genetic markers, and response to prior therapy. [2]
- Existing treatment approach usually involves use of multi-drug therapy regimens, referred to as doublet-, triplet-, or quad-therapy, with medications such as steroids, cytotoxic chemotherapy, immunomodulators (IMIDs), proteosome inhibitors (PIs), and monoclonal antibodies (mAbs). However, there is insufficient evidence to establish the safety or efficacy of the use of combination of monoclonal antibodies for MM.

- The evidence for some medications for MM is limited to the relapsed/refractory (r/r) setting, such as with the proteosome inhibitor, Ninlaro (ixazomib), as detailed in the coverage criteria. Similarly, Tecvayli (teclistamab-cqyv) is only coverable as a monotherapy for MM refractory to multiple other therapies.
- There is insufficient evidence that any one medication for MM (by mechanism of action/class) is superior to another. There is limited comparative efficacy for MM.

Pomalyst (pomalidomide) for MM

- Approval of pomalidomide was based on a single, open-label trial in 221 patients with r/r MM, comparing pomalidomide alone vs. pomalidomide plus low-dose dexamethasone. [3, 4]
  * Patients enrolled in the trial had a minimum of two prior MM therapies. Prior therapies must have included lenalidomide, and bortezomib.
  * ORRs were 7.4% and 29.2% with pomalidomide and pomalidomide plus dexamethasone, respectively.
- A second open-label trial compared pomalidomide plus low-dose dexamethasone versus high-dose dexamethasone in 455 refractory MM patients. Subjects in the pomalidomide group had a longer median overall survival at 15.4 months of follow-up (13.1 vs. 8.1 months, HR 0.75, p = 0.009). Confidence in this result is reduced by high attrition rate during the trial, the lack of blinding, and crossover between treatment groups. [5]

Kyprolis (carfilzomib) for MM

- Initial approval of carfilzomib was based on one single-arm trial in 266 subjects that evaluated ORR in patients with relapsed MM. [6]
  * Patients enrolled in the trial had received at least two prior therapies (including bortezomib and an immunomodulator [IMID], lenalidomide or thalidomide).
  * The median number of prior therapies was five and 95% were refractory to their last line of therapy.
  * The study reported an ORR of 23.7% (17.7% partial responses, 4.9% very good partial response, and 0.4% complete response).
  * There is low confidence in the evidence from the study because a cause effect relationship cannot be established due to the lack of comparator.
- A single large, randomized, open-label trial evaluated the combination of carfilzomib plus lenalidomide/dexamethasone vs. lenalidomide/dexamethasone (control group) in relapsed MM. Subjects enrolled in the trial had between one and three prior therapies. [7]
  * The median PFS was 26.3 months and 17.6 months in the carfilzomib and control arm, respectively. Corresponding ORRs were 87.1 and 66.7%.
  * Although a lower HR for death was observed in the treatment group (HR 0.79, p = 0.04), survival data is not mature; the durability and clinical meaningfulness of this difference is not fully elucidated.
  * There is low confidence in these data due to high attrition (~30%) and lack of blinding.
- Carfilzomib/dexamethasone was shown to improve median OS relative to bortezomib/dexamethasone as a subsequent therapy for r/r MM. The clinical relevance of this finding is uncertain as the majority of patients had relapsed after prior bortezomib,

which likely disadvantages the bortezomib treatment arm. However, it is supportive of the efficacy of carfilzomib as a subsequent proteosome inhibitor therapy for MM. [8]

Ninlaro (ixazomib) for MM

- The evidence for efficacy for ixazomib comes from a single randomized, controlled trial that demonstrated improvements in PFS, a surrogate endpoint, in patients who had r/r MM who had received at least one prior line of therapy. [9, 10]
    - All patients had at least one prior therapy for MM. A majority had prior bortezomib or melphalan. Patients refractory to lenalidomide or proteosome inhibitors were excluded.
    - Ixazomib improved PFS vs. lenalidomide/dexamethasone alone (20.6 vs 14.7 months).
    - A more recent RCT evaluated ixazomib as a maintenance therapy after hematopoietic stem cell transplant (HSCT). [11] An interim analysis reported a PFS advantage relative to placebo; however, OS data are not mature. The trial remains blinded and is ongoing. Ixazomib is not currently approved for use in this treatment setting.

Farydak (panobinostat) for MM

- Farydak (panobinostat) was a histone deacetylase inhibitor that received accelerated approval for relapsed or refractory MM. It was withdrawn from the US commercial market in early 2022. [12, 13, 14]
- The manufacturer is making Farydak (panobinostat) available to certain MM patients in the US who are currently receiving it. Refer to www.farydak.com/notice/ for more information.

Darzalex (daratumumab) for MM

- Darzalex (daratumumab) and Darzalex Faspro (daratumumab and hyaluronidase-fihj) are CD38-directed monoclonal antibody products for the treatment of MM. Darzalex is given intravenously (IV), while Darzalex Faspro is given subcutaneously (SC). [15, 16]
- Darzalex (daratumumab) demonstrated improved overall survival (OS) when added to a backbone regimen of either Revlimid (lenalidomide) plus dexamethasone (Rd) or Velcade (bortezomib) plus dexamethasone (Vd) in patients with relapsed or refractory MM relative to either Rd or Vd alone.
- Although there may be differences in FDA-approved indications between products, the guidelines support interchangeability. There are ongoing studies evaluating Darzalex Faspro (daratumumab and hyaluronidase-fihj) in combination with other MM medications (other than those regimens that have already been approved as safe and effective). The NCCN MM guideline footnotes Darzalex Faspro (daratumumab and hyaluronidase-fihj) as being alternative to Darzalex (daratumumab) across all of the settings for which Darzalex (daratumumab) is recommended. [2]
- Darzalex (daratumumab) is given in a dose of 16 mg/kg IV, whereas Darzalex Faspro (daratumumab and hyaluronidase-fihj) is given in a dose of 1,800 mg – 30,000 units SC. [15, 16]

Darzalex (daratumumab) intravenous (IV) formulation:

As monotherapy for relapsed and/or refractory MM:

- The efficacy of Darzalex (daratumumab) IV is based on two single-arm, unblinded clinical studies in patients with r/r MM. Patients had received a median of 4-5 prior lines of therapy. [17, 18]
    - Common prior therapies included Velcade (bortezomib), Revlimid (lenalidomide), and Pomalyst (pomalidomide).
    - A majority of patients were refractory to both a proteosome inhibitor (PI) [such as Velcade (bortezomib)] and an immunomodulatory agent (iMiD) [e.g., Revlimid (lenalidomide)].
    - Efficacy was evaluated based on ORR. The ORR in one study was 29.2% and 36% in the second study.

As an add-on to standard therapy for relapsed and/or refractory MM:

- Two RCTs evaluated Darzalex (daratumumab) IV as an add-on to backbone therapy with either Revlimid (lenalidomide) plus dexamethasone (Rd), or Velcade (bortezomib) plus dexamethasone (Vd) in patients with r/r MM who had at least one prior therapy. [20, 21]
    - In a final analysis of these two trials a significant improvement in OS was reported with add-on Darzalex (daratumumab) relative to Rd or Vd alone. [50, 51]
    - The following flaws may lower confidence in the reported results: Neither study was blinded, and performance bias due to high attrition cannot be ruled out.
- An uncontrolled (single-arm), open-label trial evaluated Darzalex (daratumumab) IV as an add-on to backbone therapy with dexamethasone plus Pomalyst (pomalidomide). Patients enrolled in the study had a median of four prior MM therapies. [22]
    - The ORR was 59.2%, with 5.8% complete responses. Although results may appear impressive relative to historical controls, it cannot be concluded that add-on Darzalex (daratumumab) IV improves any clinical outcome relative to dexamethasone and Pomalyst (pomalidomide) alone.
    - Evidence from this trial is of very low quality due to the lack of comparator and use of an unvalidated surrogate endpoint.

As a primary MM therapy when autologous stem cell transplant is not an option:

- A randomized, open-label trial showed improved PFS at 18 months with Velcade (bortezomib)/melphalan/prednisone (BMP) plus Darzalex (daratumumab) IV vs. BMP alone, in newly diagnosed MM not eligible for an autologous stem cell transplant (ASCT). [23]
    - Patients enrolled in the trial either had coexisting conditions which precluded them from receiving high-dose chemotherapy with ASCT, or were 65 years of age or older (92% of the population).
    - The 18-month PFS was 71.6% [95% CI, 65.5, 76.8] and 50.2% [43.2, 56.7%] in the daratumumab and control groups, respectively. Median follow up at the time of the interim analysis was 16.5 months.

Darzalex Faspro (daratumumab and hyaluronidase-fihj) subcutaneous (SC) formulation: [16]

- The safety and efficacy of Darzalex Faspro (daratumumab and hyaluronidase-fihj) SC is based on previous Darzalex (daratumumab) IV studies. Pharmacokinetic studies and small, single-arm, observational trials evaluating PFS in the following settings were used as confirmatory evidence for Darzalex Faspro (daratumumab and hyaluronidase-fihj):
    - Newly diagnosed MM in combination with Velcade (bortezomib), melphalan, and prednisone.
    - r/r MM in combination with Revlimid (lenalidomide) and dexamethasone.
    - r/r MM as a monotherapy after disease progression on at least three prior therapies (including a proteosome inhibitor and immunomodulator), or progression after at least two prior PI and iMiD combination regimens.

Sarclisa (isatuximab-irfc) for MM

- Sarclisa (isatuximab-irfc) is an intravenously administered CD38-directed monoclonal antibody. Its mechanism of action is similar to that of Darzalex (daratumumab) IV.
- The initial FDA approval was based on low-quality data from a single, open-label (non-blinded), randomized, controlled trial in r/r MM. The trial compared Sarclisa (isatuximab-irfc), Pomalyst (pomalidomide) and dexamethasone (IPD) with PD alone. The overall confidence in the study results is limited due to several concerns (lack of blinding, potential imbalance in populations, and high rate of differential attrition). [24, 25]
    - Patients had been treated with a minimum of two prior MM therapies.
    - All patients had no response to prior Revlimid (lenalidomide) and proteosome inhibitors (used either separately or in combination). Non-response was defined as disease progression on or within 60 days, intolerance to Revlimid (lenalidomide) or the proteosome inhibitor, or disease progression within 6 months after achieving at least a partial response.
    - Patients with prior use of Pomalyst (pomalidomide) were not allowed in the study, nor were those with disease refractory to prior therapy with daratumumab, another CD-38 monoclonal antibody. *Refractory was defined as having achieved an initial response with subsequent disease progression while on therapy, or progression within 60 days of the last dose.*
    - The trial reported a PFS benefit with the addition of Sarclisa (isatuximab-irfc) to PD, a surrogate endpoint. OS will be analyzed as a secondary endpoint; however, no significant difference in survival between the study arms has been noted to date.

Empliciti (elotuzumab) for MM

- Empliciti (elotuzumab) is an intravenously administered SLAMF7-directed immunostimulatory monoclonal antibody for the treatment of r/r MM.
- The initial evidence for efficacy of Empliciti (elotuzumab) was based on a single, phase 3, randomized, open-label trial in patients who had received one to three prior therapies for MM [ELOQUENT-2 (NCT01239797)]. [26] The median number of prior treatments was two. Velcade (bortezomib) was the most common prior therapy (70%), followed by melphalan (65%), Thalomid (thalidomide) (48%), and Revlimid (lenalidomide) (6%).

Empliciti (elotuzumab) improved PFS, a surrogate endpoint. However, the effect of Empliciti (elotuzumab) on clinically relevant outcomes such as overall survival or quality of life is not known.

* Empliciti (elotuzumab) plus Revlimid (lenalidomide)/dexamethasone was compared to Revlimid (lenalidomide)/dexamethasone alone.
* Elotuzumab resulted in a 4.5-month PFS advantage compared to Revlimid (lenalidomide) and dexamethasone alone (19.4 months vs 14.9 months, respectively).

- Subsequently, Empliciti (elotuzumab) was studied in combination with Pomalyst (pomalidomide)/dexamethasone vs. Pomalyst (pomalidomide)/dexamethasone in a single, phase 3, randomized, open-label trial (n=117) in patients with r/r MM [ELOQUENT-3]. [27] Empliciti (elotuzumab) improved ORR, as well as PFS, surrogate endpoints. However, the effect of Empliciti (elotuzumab) on clinically relevant outcomes such as overall survival or quality of life is not known.
  * The median number of prior treatments was three. Prior therapies included stem cell transplant (55%), Velcade (bortezomib) (100%), Revlimid (lenalidomide) (99%), cyclophosphamide (66%), melphalan (63%), Kyprolis (carfilzomib) (21%), and Darzalex (daratumumab) IV (3%).
  * The patient population was highly-refractory to prior therapies: Revlimid (lenalidomide)-refractory (87%); proteosome inhibitor-refractory (80%); Revlimid (lenalidomide)- and proteosome inhibitor-refractory (70%).
  * Empliciti (elotuzumab) improved PFS by 5.58-months vs. Pomalyst (pomalidomide)/dexamethasone alone (10.25 months vs 4.67 months, respectively).
- A smaller, preliminary (phase 2) trial evaluated elotuzumab as an add-on to Velcade (bortezomib) plus dexamethasone. [28, 29] A 2.8-month PFS advantage was reported.

Blenrep (belantamab mafodotin-blmf) for MM [30, 38]

- Blenrep (belantamab mafodotin-blmf) was recently withdrawn from the market and is no longer available. The reason for withdrawal is failure to show a clinical benefit over standard therapy in the trial intended to confirm efficacy for regular FDA approval.
- Patients who are already enrolled in the Blenrep REMS program will have the option to enroll in a Compassionate use program through the manufacturer. (www.blenrep.com)

Pepaxto (melphalan flufenamide) for MM

- Pepaxto (melphalan flufenamide) has been withdrawn from the market because harms were found to exceed any potential for benefit in MM.

Tecvayli (teclistamab-cqyv) and Elrexfio (elranatamab-bcmm) for MM [47-49, 54, 55]

- Tecvayli (teclistamab-cqyv) and Elrexfio (elranatamab-bcmm) are intravenously administered bispecific B-cell maturation antigen (BCMA)-directed T-cell engagers for the treatment of multi-refractory MM.
- Tecvayli (teclistamab-cqyv) and Elrexfio (elranatamab-bcmm) each received FDA accelerated approval based on small, single-arm observational studies that evaluated objective response rate (ORR) in patients with triple class exposed (immunomodulator,

proteosome inhibitor, CD38-directed monoclonal antibody) patients with relapsed or refractory MM.

 * Patients in each of the studies had a median of five prior therapies.
 * The studies evaluated tumor response as the primary surrogate endpoint:
 o *Tecvayli (teclistamab-cqyv):* The ORR was 62%, with 28% of subjects having a complete response or better.
 o Elrexfio (elranatamab-bcmm): The ORR was 58%, with 26% of subjects having a complete response or better.

- It has not been determined whether Tecvayli (teclistamab-cqyv) or Elrexfio (elranatamab-bcmm) improve any clinically important outcomes (e.g., overall survival), or whether the potential for benefit is greater than the risks of therapy.
- Access to both Tecvayli (teclistamab-cqyv) and Elrexfio (elranatamab-bcmm) is restricted via Risk Evaluation and Mitigation Strategies (REMS).

Talvey (talquetamab-tgvs) for MM [52, 53]

- Talvey (talquetamab-tgvs) is an intravenously administered bispecific GPRC5D-directed T-cell engager for the treatment of MM.
- Talvey (talquetamab-tgvs) received FDA accelerated approval based on a small, single-arm observational study that evaluated objective response rate (ORR) in patients with triple class exposed (immunomodulator, proteosome inhibitor, CD38-directed monoclonal antibody) patients with relapsed or refractory MM.
 * Patients enrolled in the study had a median of five prior MM therapies.
 * The ORR was 73%, with 35% of subjects having a complete response or better.
- It has not been determined whether Talvey (talquetamab-tgvs) improves any clinically important outcome (e.g., overall survival) or whether the potential for benefit is greater than the risks of therapy.
- Access to Talvey (talquetamab-tgvs) is restricted via the Talvey Risk Evaluation and Mitigation Strategy (REMS).

Kaposi Sarcoma (KS)

- The approval of Pomalyst (pomalidomide) in KS is based on an observational trial that followed 18 patients who were HIV-positive, and 10 patients who were HIV-negative. The trial evaluated tumor response as the endpoint. [35]
- The trial excluded patients with symptomatic pulmonary or visceral KS.
- The majority (75%) of subjects in the trial had received prior chemotherapy for their KS. Patients who were HIV-positive also had to have been receiving highly active anti-retroviral therapy (HAART) for a minimum of two to three months without regression of their KS prior to being treated with pomalidomide. [35]
- The NCCN AIDS-related KS guideline recommends liposomal doxorubicin as the preferred front-line therapy for KS. Treatment in HIV-negative patients parallels treatment in the HIV-positive population. Pomalidomide is a preferred regimen after failure of front-line chemotherapy. [36]

Light chain amyloidosis (AL)

- The approval (FDA Accelerated approval) of Darzalex Faspro (daratumumab and hyaluronidase-fihj) SC in light chain amyloidosis (AL) is based on an open-label RCT [ANDROMEDA] that evaluated an unvalidated surrogate as the primary endpoint. As per FDA Accelerated approval regulations, additional evidence is needed to confirm there is a clinical benefit for continued approval. [16]
  * The trial compared the addition of Darzalex Faspro to Velcade (bortezomib)/ cyclophosphamide/dexamethasone (D-VCd) with VCd alone.
  * Patients enrolled in the study had newly diagnosed, measurable (hematologic) disease that affected at least one organ (e.g., heart, liver, kidney).
  * Patients with certain cardiac disease (e.g., NYHA Class IIIB and IV) where not allowed to enroll in the trial.
  * Per protocol, therapy was administered for a maximum of two years, or until there was disease progression.
  * Complete hematologic response HemCR, the primary endpoint, was achieved in 42% and 13% of the D-VCd and VCd treatment groups, respectively.
  * HemCR has not been shown to accurately predict any clinically important outcome (e.g., improved survival or quality of life).
- The NCCN Systemic Light Chain Amyloidosis guideline lists D-VCd among the preferred treatment options for front-line use in AL. [44]

Safety [37]

- Overall, medications for MM, by class/mechanism of action, appear to have similar safety profiles (based on indirect comparisons). There is no reliable evidence to allow conclusion that any one medication for MM (by class) is safer or more tolerable that another, given lack of comparative safety data. Common adverse events include:
  * IMIDs: Neutropenia, fatigue, anemia, constipation, nausea, diarrhea, dyspnea, upper respiratory tract infections, back pain, and pyrexia. Boxed warnings for embryo-fetal toxicity, venous thromboembolism, and hematologic toxicity. A restricted distribution program (Risk Evaluation and Management Strategy, "REMS") is in place to prevent accidental fetal exposure. Doses may be modified for hematologic toxicity.
  * Proteosome inhibitors: gastrointestinal toxicity, thrombocytopenia, and peripheral neuropathy; severe: cardiac toxicity.
  * Monoclonal antibodies: infusion-related reactions, neutropenia, upper respiratory tract infections, and diarrhea. Premedication with antihistamines, antipyretics, and corticosteroids is recommended. Of note: The safety of Darzalex Faspro (daratumumab and hyaluronidase-fihj) parallels that of the intravenous product.
  * Alkylating Agents: bone marrow suppression (including anemia, leukopenia, lymphocytopenia, neutropenia, and thrombocytopenia), gastrointestinal toxicity (including nausea and vomiting), renal toxicity, fatigue. Boxed warnings for severe marrow suppression leading to infection or bleeding, hypersensitivity reactions (including anaphylaxis), and potential secondary malignancy.

- Overall tolerability of some medications for MM may limit utility:
  * HDAC inhibitor Farydak (panobinostat): labeling contains a boxed warning for severe diarrhea (25% severe), and cardiac toxicity, including severe and fatal cardiac ischemic events, as well as severe arrhythmias. Therapy should be interrupted at the onset of moderate diarrhea (4 to 6 stools per day). Other serious toxicities include hemorrhage, hepatotoxicity, and embryo-fetal toxicity. Electrolyte abnormalities are also common. [13]
  * Tecvayli (teclistamab-cqyv): labeling contains a boxed warning for cytokine release syndrome (CRS) and neurological toxicity, including Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS). Use is restricted through the Tecvayli Risk Evaluation and Mitigation Strategy (REMS). [47-49]
  * Xpovio (selinexor): is associated with significant toxicity, need for dose modifications, and treatment-related deaths (see Xpovio policy for details). [39-42]

Appendix 1: Classification of Medications used for Multiple Myeloma		
Chemotherapy	**Immunomodulators (IMIDs)**	**Proteosome Inhibitors (PIs)**
- cyclophosphamide - doxorubicin - Doxil (liposomal doxorubicin) - melphalan HCl (generic, Evomela) - vincristine	- Revlimid (lenalidomide) - Pomalyst (pomalidomide) - Thalomid (thalidomide)	- bortezomib (generic Velcade) - Kyprolis (carfilzomib) - Ninlaro (ixazomib)
Exportin 1 inhibitor	**Anti-BCMA therapy**	**Monoclonal Antibodies (mAbs)**
- Xpovio (selinexor)	- Abecma (idecabtagene vicleucel) - Carvykti (ciltacabtagene autoleucel) - Elrexfio (elranatamab-bcmm) - Tecvayli (teclistamab-cqyv)	- Anti-CD38 - Darzalex (daratumumab) - Sarclisa (isatuximab-irfc) - Anti-SLAMF7 - Empliciti (elotuzumab)
GPRC5D-Directed therapy		
- Talvey (talquetamab-tgvs)	-	-

Table 3. Investigational Uses	
There is insufficient evidence to establish the safety or efficacy of the medications for MM (as listed below) for the treatment of the listed conditions. Additional studies are ongoing for many of these conditions. [43]	
Front-line treatment of MM	Empliciti (elotuzumab), Ninlaro (ixazomib), and Sarclisa (isatuximab-irfc) are being studied in the front-line multiple myeloma setting. However, the evidence is considered preliminary.
Sequential use of BCMA-directed agents	There is unpublished, low-quality evidence suggesting Elrexfio (elranatamab) may have some activity in patients who had prior exposure to BCMA-directed therapy; however, data is limited to tumor responses in small numbers of patients. Additional information is needed to establish efficacy in these patients. Sequential use of BCMA-directed therapies remains investigational.
Myelofibrosis	There are several small, published studies that evaluate the use of Pomalyst (pomalidomide) in patients with myelofibrosis. Several other medications for MM are also being studied. The evidence is considered insufficient at this time.
Smoldering multiple myeloma	There are several ongoing studies to evaluate various MM therapies [Kyprolis (carfilzomib), Darzalex (daratumumab), Darzalex Faspro (daratumumab and hyaluronidase-fihj), Empliciti (elotuzumab), Ninlaro (ixazomib), and Sarclisa (isatuximab-irfc)] in smoldering plasma cell myeloma. The evidence is considered insufficient at this time.
Systemic light chain amyloidosis (AL), in the absence of multiple myeloma	- Pomalyst (pomalidomide) in combination with dexamethasone is one of many potential treatment regimens listed in the NCCN Systemic Light Chain Amyloidosis guidelines; however, the guidelines state that optimal therapy for systemic light chain amyloidosis remains unknown and treatment in the context of a clinical trial is strongly encouraged when possible. Other listed treatment options include, but are not limited to, cyclophosphamide/thalidomide/dexamethasone, dexamethasone/alpha-interferon, oral melphalan/dexamethasone, and thalidomide/dexamethasone. [44] - There are several ongoing studies listed in clinicaltrials.gov that are designed to evaluate Sarclisa (isatuximab-irfc) in other disease settings including amyloidosis. There are no results posted to date.
Waldenström's macroglobulinemia	- There are ongoing trials that are studying Pomalyst (pomalidomide) and Kyprolis (carfilzomib) in Waldenström's macroglobulinemia. There is no published data supporting the safety and efficacy of Pomalyst (pomalidomide) in these populations. - There is currently insufficient evidence to support the use of Kyprolis (carfilzomib) in combination with rituximab and dexamethasone for the treatment of Waldenström's macroglobulinemia. Although Kyprolis (carfilzomib)/rituximab/dexamethasone is listed in NCCN guidelines as a category 2A recommendation, the only information to date consists of a single-center, uncontrolled phase 2 study in 31 patients. [45] Well-designed studies are necessary to establish efficacy and benefit in these populations.
Solid tumors	There are also several ongoing trials that are studying Pomalyst (pomalidomide) in soft tissue sarcoma. There is no published data supporting the safety and efficacy of Pomalyst (pomalidomide) in these populations.

Cross References
Products with Therapeutically Equivalent Biosimilars/Reference Products, Medication Policy Manual, Policy No. dru620
Chimeric Antigen Receptor (CAR) T-cell Therapies, Medication Policy Manual, Policy No. dru523
Xpovio, selinexor, Medication Policy Manual, Policy No. dru607

Codes	Number	Description
HCPCS	J9145	Injection, daratumumab (Darzalex), 10 mg
HCPCS	J9144	Injection, daratumumab and hyaluronidase-fihj (Darzalex Faspro), 10 mg
HCPCS	J9176	Injection, elotuzumab (Empliciti), 1 mg
HCPCS	J1323	Injection, elranatamab-bcmm (Elrexfio), 1 mg
HCPCS	J9047	Injection, carfilzomib (Kyprolis), 1 mg
HCPCS	J9227	Injection, isatuximab-irfc (Sarclisa), 10 mg
HCPCS	J9380	Injection, teclistamab-cqyv (Tecvayli), 0.5 mg
HCPCS	J3055	Injection, talquetamab-tgvs (Talvey), 0.25 mg

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Revision History

Revision Date	Revision Summary
6/20/2024	No criteria changes with this annual review.
3/21/2024	No criteria changes with this annual review.
12/7/2023	Coverage for Elrexfio (elranatamab-bcmm) was added to the policy when used as a monotherapy for patients with MM who have had at least four prior MM regimens which must have included an anti-CD38 mAb, a proteosome inhibitor, and an immunomodulatory agent when there has been no prior therapy with therapy directed against BCMA. Coverage for Talvey (talquetamab-tgvs) was added to the policy when used as a monotherapy for patients with MM who have had at least four prior MM regimens which must have included an anti-CD38 mAb, a proteosome inhibitor, and an immunomodulatory agent.
6/15/2023	No changes to policy criteria with this annual update.
3/16/2023	• Blenrep (belantamab mafodotin) was removed from the policy because the manufacturer has ceased marketing of this product in the U.S. • Coverage for Tecvayli (teclistamab-cqyv) was added to the policy when used as a monotherapy for patients with MM who have had at least four prior MM regimens which must have included an anti-CD38 mAb, a proteosome inhibitor, and an immunomodulatory agent when there has been no prior therapy with therapy directed against BCMA.
6/17/2022	Effective 9/1/2022: • Farydak (panobinostat) was removed from this policy because the manufacturer has ceased marketing of this product in the US. • Pepaxto (melphalan flufenamide) removed from policy due to withdrawal from the market. *Note: Revisions were made to update to current standard policy language; however, there was no change to the intent of this policy.*
10/15/2021	Pepaxto (melphalan flufenamide) changed to investigational for all indications due to a safety signal that indicates numerically higher incidence of death with melphalan flufenamide versus standard of care.
7/16/2021	New combination policy (effective 10/1/2021): • The individual Multiple Myeloma policies were combined, with the exception of selinexor (Xpovio, dru607) and idecabtagene vicleucel (Abecma; a newer CAR-T therapy, dru523). • Revlimid will no longer require pre-authorization. • The newly FDA approved Pepaxto (melphalan flufenamide) added to policy. The use of Pepaxto for multiple myeloma will be considered not medically necessary and therefore not covered due to lack of proven additional benefit versus lower-cost intravenous melphalan HCl. • Added coverage criteria for Darzalex (daratumumab), Darzalex Faspro (daratumumab and hyaluronidase-fihj) in systemic light chain amyloidosis, a newly FDA approved indication.

Revision Date	Revision Summary
	• Coverage criteria now explicitly list that combination of any two monoclonal antibodies (Darzalex, Empliciti, Sarclisa; per Appendix 1) is not coverable. No change to intent. • Investigational Uses were simplified (any use without coverage criteria is "Investigational"). • Clarification of quantity limits, for operational consistency. • Removed metastatic castration-resistant prostate cancer (for lenalidomide) from the list of 'Not Medically Necessary' uses (this indication was also listed under 'Investigational Uses').

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